
ATTENTION TEMPORAL CONVOLUTIONAL NETWORK WITH WEIGHT SHARING FOR MOTOR IMAGERY SIGNAL CLASSIFICATION

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ABSTRACT

Brain-computer interfaces (BCIs) offer a non-muscular communication pathway by enabling users to interact with external devices through brain signals alone. A widely adopted BCI paradigm is based on motor imagery (MI), where individuals imagine specific limb movements without actual muscle activity. Electroencephalography (EEG) is commonly employed to capture these MI-induced neural signals, which are then processed and classified to infer the user's intended action. However, the inherent low signal-to-noise ratio of EEG data, along with significant inter-subject variability, poses substantial challenges for accurate and robust classification. In this work, we propose the Attention Temporal Convolutional Network with Weight Sharing (ATCNet-WS) to enhance MI EEG classification performance. ATCNet-WS employs multi-head self-attention to capture discriminative spatio-temporal features and utilizes multi-layer dilated temporal convolutions to aggregate information across long temporal contexts. A sliding-window based multi-branch architecture is applied to provide additional network width. To mitigate overfitting, weight sharing is applied across temporal convolution layers with varying dilation rates. Cosine annealing learning rate schedule with warm restarts is adopted to accommodate the different training phases effectively. ATCNet-WS achieves subject-dependent four-class classification accuracy of 87.06% on the BCI Competition IV 2a dataset, and subject-dependent two-class classification accuracy of 87.85% on the BCI Competition IV 2b dataset, both outperforming state-of-the-art methods.

Keywords Weight sharing · bagging · cosine annealing · motor imagery (MI) · electroencephalography (EEG) · classification

1 Introduction

Brain-Computer Interfaces (BCIs) represent a class of neurotechnologies that establish a direct communication pathway between the brain and external devices, such as computers or robotic systems [1]. By decoding neural activity into actionable commands, BCIs bypass conventional neuromuscular output pathways—enabling individuals with severe motor impairments to interact with their environment. Applications of BCIs include control of assistive devices (e.g., wheelchairs or robotic prostheses) [2], restoration of communication in locked-in patients [3], and treatment of depression or post-traumatic stress disorder (PTSD) through neurofeedback mechanisms [4].

BCIs are broadly categorized into invasive and non-invasive systems. Invasive BCIs involve the surgical implantation of electrodes directly into cortical tissue, offering high spatial and temporal resolution at the cost of increased surgical and long-term health risks. In contrast, non-invasive BCIs employ external sensors to monitor brain activity without the need for surgery. Although they provide lower signal fidelity due to noise introduced by the skull and scalp, non-invasive systems are safer, more accessible, and suitable for widespread clinical use.

Among non-invasive BCIs, electroencephalography (EEG)-based BCIs are the most widely utilized. EEG captures electrical potentials generated by neuronal ensembles using electrodes placed on the scalp. These signals are typically classified into distinct frequency bands—such as alpha (8–13 Hz), beta (13–30 Hz), and theta (4–7 Hz)—each associated with different cognitive or physiological states. By analyzing these spectral features, EEG-based BCIs can infer user intent and translate it into control commands for external devices.

Motor imagery (MI) refers to the mental simulation of movement without any actual motor execution [5]. During MI, distinct changes occur in the brain’s electrical activity, particularly in the sensorimotor cortex. These changes show as either increases or decreases in the amplitude of rhythmic EEG components, known respectively as event-related synchronization (ERS) and event-related desynchronization (ERD) [6]. The terminology stems from the neurophysiological basis of EEG signals: signals reflect the synchronous firing of large populations of neurons within a localized region, resulting in increased signal amplitude. Conversely, desynchronization leads to attenuated EEG signal strength due to phase cancellation between opposing potentials.

Distinct MI tasks—such as imagining movements of the left hand versus the right hand—engage different cortical areas and produce characteristic ERD/ERS patterns [6] [7]. These patterns form the foundation for decoding user intent in MI-based BCIs. By accurately classifying MI-induced EEG features, BCIs can interpret user commands to control external devices. However, the low signal-to-noise ratio of non-invasive EEG poses significant challenges. Discriminating between MI states or separating MI from non-MI cognitive activities remains difficult. Consequently, robust signal processing techniques and advanced classification algorithms are essential to enhance decoding performance. Continued research in feature extraction [8], spatial filtering, and deep learning [9] approaches is critical to advancing the reliability and practical utility of MI-based BCIs.

In traditional MI classification approaches, the Common Spatial Pattern (CSP) algorithm [10] remains one of the most widely adopted and foundational methods. CSP seeks optimal spatial filters performing linear operations across EEG channels to maximize the variance difference between two classes. Building upon CSP, several variants have been proposed to enhance classification performance, including Filter Bank CSP (FBCSP) [11], Probabilistic CSP (P-CSP) [12], and Discriminative FBCSP (DFBCSP) [13]. Brodu et al [14] compared multiple feature extraction methods and examined their combinations to determine the configuration yielding optimal performance. Park et al [15] presented an improved CSP that, instead of utilizing all EEG channels, selected the most informative cortical regions and extracted features from a subset of local channels. Gaur et al [16] introduced a strategy based on sliding window to further strengthen CSP performance. These methods typically follow a standard pipeline: features are extracted using CSP-based algorithms, relevant features are selected using criteria such as the Fisher score [17], and classification is performed using techniques such as Linear Discriminant Analysis (LDA). This framework has become the conventional paradigm for MI classification.

Traditional methods heavily rely on handcrafted features and manually designed computing procedures, which limit their performance. In recent years, deep learning has emerged as a dominant approach in MI classification, offering the ability to automatically learn discriminative features directly from raw EEG data. Early deep learning models such as EEGNet [18] and ShallowConvNet [19] adopted Convolutional Neural Network (CNN) architectures, where convolutional layers were designed to replace individual steps in traditional pipelines (e.g., FBCSP) to enable end-to-end feature extraction and classification during training.

Subsequent developments have extended these foundational models. Ingolfsson et al. [20] proposed the Temporal Convolutional Network (TCNet), which augments EEGNet by incorporating temporal convolution layers to better capture temporal dynamics. Building upon TCNet, Musallam et al. [21] introduced various enhancements that allow the model to achieve high classification accuracy across different subjects using a single set of hyperparameters. The Multi-Branch EEGNet with Squeeze-and-Excitation blocks (MBEEGSE) [22] further extends EEGNet by integrating multi-branch pathways and channel-wise attention mechanisms to improve performance.

Kim et al. [23] identified structural limitations in ShallowConvNet and substantially improved its accuracy by modifying the layer order, adjusting network hyperparameters, and optimizing the training procedure. Dai et al [24] developed a CNN architecture with multiple convolutional scales to capture features at different receptive field sizes. Amin et al [25] integrated features extracted from networks with varying depths and convolutional scales to improve discriminative capability. The Multi-Scale Vision Transformer Neural Network (MSVTNet) [26] employs multiple parallel branches in the early layers, each utilizing different filter scales to capture diverse feature representations. The Attention Temporal Convolutional Network (ATCNet) [27] enhances TCNet by incorporating attention mechanisms and a sliding window approach, leading to significant performance gains. Building upon this, the Dual-Branch ATCNet (DB-ATCNet) [28] introduces a two-branch convolutional block structure and integrates additional attention layers to further refine feature extraction and classification. Zhang et al [29] proposed EEG-Inception, an architecture composed of multiple inception modules for multi-scale EEG feature representation. Transfer learning has been leveraged to improve model generalization. Zhang et al [30] proposed a fine-tuning strategy in which different layers of a pretrained

network were frozen, with the optimal configuration determined by classification accuracy. Kant et al [31] applied the Continuous Wavelet Transform (CWT) to convert one-dimensional EEG signals into two-dimensional time–frequency representations, enabling transfer learning models to more effectively exploit spatial–temporal features.

In this paper, we propose the Attention Temporal Convolutional Network with Weight Sharing (ATCNet-WS) for MI EEG decoding. ATCNet-WS is evaluated on the BCI Competition IV 2a and 2b [32] datasets and achieves an average classification accuracy of 87.06% on 2a dataset and 87.85% on 2b dataset, which, to the best of our knowledge, represents the highest reported performance on these datasets to date.

The main contributions of this paper are summarized as follows:

1. We propose a novel model, ATCNet-WS, to address the MI classification task. ATCNet-WS builds upon the powerful sliding window, multi-head attention, and multi-layer dilated temporal convolution architectures.
2. Sharing weights between temporal convolution layers with different dilation rates reduces model complexity, thus mitigates overfitting.
3. Extension of the input signal increases the number of sliding windows, enabling the model to leverage richer temporal information.
4. A bagging ensemble strategy is adopted. We introduce a simple yet effective sample replication technique, which, when combined with bagging, demonstrates enhanced performance.
5. A two-phase cosine annealing learning rate schedule is applied to optimize training dynamics and escape local minima.
6. ATCNet-WS achieves excellent performance on the BCI Competition IV 2a and 2b datasets. The source code for ATCNet-WS is publicly available at: <https://github.com/Xuzefeng131/WS>.

The remainder of this paper is organized as follows. Section II provides a detailed description of the proposed ATCNet-WS architecture. Section III presents the experimental results, including performance comparison against models from the literature and an ablation analysis to evaluate the contribution of each proposed enhancement. Finally, Section IV concludes the paper and discusses potential directions for future work.

2 Methods

This section provides a detailed description of the proposed ATCNet-WS model. As illustrated in Fig. 1, the ATCNet-WS architecture consists of four main components: a convolutional block, a sliding window block, an attention and temporal convolution block, and a decision block. The convolutional block, inspired by the structure of EEGNet, comprises three convolutional layers that perform signal filtering and inter-channel linear operations. The sliding window block segments the input EEG signal into multiple overlapping windows. Each window is then processed independently through an attention and temporal convolution block branch. The outputs of all branches are aggregated in the decision block to produce the final classification output.

2.1 Dataset and Preprocessing

We evaluated the performance of the proposed algorithm using the BCI Competition IV datasets 2a and 2b, which are widely adopted benchmarks for MI EEG classification.

The BCI Competition IV 2a dataset consists of EEG recordings collected from 22 channels across nine subjects. Each subject completed two recording sessions on different days, designated as the training and testing sessions. Each session contains 288 trials. The EEG signals were sampled at 250 Hz and band-pass filtered between 0.5 Hz and 100 Hz. The MI tasks comprise four classes: movement of left hand, right hand, foot, and tongue.

The BCI Competition IV 2b dataset includes EEG recordings from three channels, also acquired from nine subjects. Each subject performed five sessions, with the first two sessions conducted without visual feedback and the remaining three with feedback. The number of trials per session ranges from 60 to 160. For our experiments, we used the first three sessions—two with feedback and one without—as the training set, and the remaining two feedback sessions as the testing set. The EEG signals were sampled at 250 Hz and band-pass filtered between 0.5 Hz and 100 Hz. The MI tasks comprise two classes: movement of left hand and right hand.

In both datasets, each trial is represented using a 5-second segment of EEG, 1 second before and 4 seconds after the presentation of the cue indicating the MI task to be performed, resulting in a total of 1250 time points per trial.

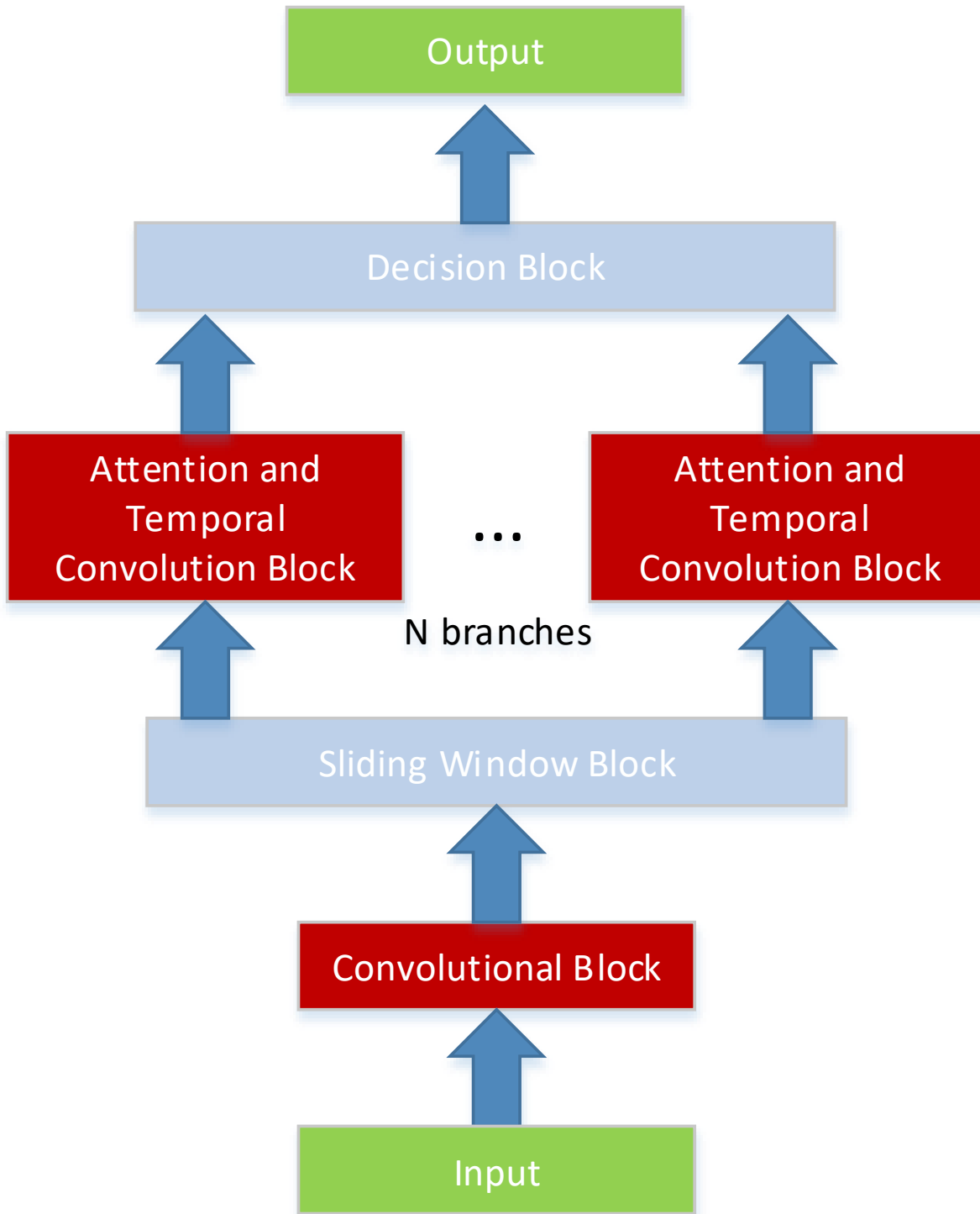


Figure 1: Overall architecture of ATCNet-WS.

2.2 Convolutional Block

The architecture of the convolutional block, adapted from EEGNet [18], is illustrated in Fig. 2. This block comprises three convolutional layers. The first convolutional layer performs temporal filtering with 16 kernels, each having a size of 1×64 . These temporal filters act similarly to filters in the FBCSP [11] algorithm. This layer is followed by batch normalization. The second convolutional layer conducts spatial filtering through depthwise convolutions across the channel dimension. Specifically, grouped convolution is applied with a kernel size of $C \times 1$, where C denotes the number of EEG channels (e.g., $C=22$ for dataset 2a and $C=3$ for dataset 2b). Each filtered signal from the first convolutional layer outputs two feature maps. This operation mimics the linear channel combination step in the CSP [10] method. Batch normalization and an Exponential Linear Unit (ELU) activation function are applied subsequently. An average pooling layer follows to reduce the temporal dimension by a factor of eight. The third convolutional layer employs 32 kernels with a size of 1×64 , further extracting abstracted temporal features. This layer is also followed by batch normalization and ELU activation. Another average pooling layer is applied afterward to reduce the temporal length by an additional factor of seven.

2.3 Sliding Window

After processing a single trial through the convolutional block, the resulting feature map has a shape of 32×22 . A sliding window of length 16 is applied along the temporal dimension (i.e., the second dimension), shifting one time step at a time. This operation yields a total of seven overlapping sub-matrices of size 32×16 , from which we utilize the last six sub-matrices for subsequent processing. Each of these sub-matrices is then independently passed through an Attention and Temporal Convolution branch, resulting in six parallel branches.

2.4 Attention and Temporal Convolution (ATC) Block

This block consists of two primary components: an attention module and a temporal module.

As shown in Fig. 3, the attention module begins with Layer Normalization, followed by a Multi-Head Self-Attention (MSA) [33]. The output of the MSA is then added to the normalized input, forming a residual connection [34]. The MSA is configured with two attention heads, each with a key dimension of 8, and includes a dropout layer with a dropout rate of 0.5 to improve generalization.

The temporal module comprises two residual blocks, as shown in Algorithm 1, each containing a pair of dilated causal temporal convolutional layers [20]. Each convolutional layer employs 32 kernels with a kernel size of 4. After each convolution, a sequence of Batch Normalization, ELU activation, and Dropout layers is applied. A residual connection is formed by adding the input of the residual block to its output. Both the attention and temporal modules preserve the dimensionality of the tensor, maintaining the shape of 32×16 throughout.

After the two modules, the final output of the ATC block is a 32×1 vector, obtained by extracting the features at the last time step along the temporal axis. To ensure that the receptive field spans the full temporal window of length 16, the dilation parameters of the two residual blocks are set to 1 and 2, respectively. This configuration allows the output vector to effectively aggregate information from the entire tensor.

To reduce model complexity, we introduce a weight sharing strategy, in which the two dilated temporal convolutional layers in each residual block share the same set of parameters. The detailed configurations of the temporal module is provided in Algorithm 1. As illustrated, the residual blocks contain two independently defined dilated temporal convolution layers (conv1, conv2) and each used twice in the blocks with different dilation rates.

2.5 Decision Block

We use an average mode for integrating the outputs from the branches to produce the final classification result. In this approach, the output vector from each branch (of size 32×1) is independently passed through a dense layer with T output units, where T is the number of classes, resulting in a $T \times 1$ vector per branch. These vectors are then averaged element-wise across branches to produce a single $T \times 1$ output vector, which is subsequently passed through a softmax function to yield the final class probabilities.

2.6 Bagging

Bagging is the ensemble learning technique that enhances classification accuracy by combining the predictions of multiple base models. The effectiveness of bagging largely depends on the strength and diversity of the individual models—greater difference among them generally leads to improved ensemble performance.

Algorithm 1 Temporal module of ATCNet-WS

Input: tensor x
Output: tensor y

```

1:  $conv1 \leftarrow \text{conv1D\_layer}(\text{dilation\_rate}=1)$ 
2:  $conv2 \leftarrow \text{conv1D\_layer}(\text{dilation\_rate}=1)$ 
3:  $z \leftarrow x$ 
4:  $x \leftarrow conv1(x)$ 
5:  $x \leftarrow \text{BatchNorm}(x)$ 
6:  $x \leftarrow \text{ELU}(x)$ 
7:  $x \leftarrow \text{Dropout}(x)$ 
8:  $x \leftarrow conv2(x)$ 
9:  $x \leftarrow \text{BatchNorm}(x)$ 
10:  $x \leftarrow \text{ELU}(x)$ 
11:  $x \leftarrow \text{Dropout}(x)$ 
12:  $x \leftarrow x + z$ 
13:  $x \leftarrow \text{ELU}(x)$ 
14:
15:  $conv1.\text{dilation\_rate} \leftarrow 2$ 
16:  $conv2.\text{dilation\_rate} \leftarrow 2$ 
17:  $z \leftarrow x$ 
18:  $x \leftarrow conv1(x)$ 
19:  $x \leftarrow \text{BatchNorm}(x)$ 
20:  $x \leftarrow \text{ELU}(x)$ 
21:  $x \leftarrow \text{Dropout}(x)$ 
22:  $x \leftarrow conv2(x)$ 
23:  $x \leftarrow \text{BatchNorm}(x)$ 
24:  $x \leftarrow \text{ELU}(x)$ 
25:  $x \leftarrow \text{Dropout}(x)$ 
26:  $y \leftarrow x + z$ 
27:  $y \leftarrow \text{ELU}(x)$ 
28:
29: return  $y$ 

```

In our implementation, we perform a 5-fold partitioning of the training set, and each fold is used to train a separate instance of the model. During inference, the final prediction is obtained by aggregating the outputs of the five models using a majority voting scheme.

In our experiments, we observed that duplicating the training dataset led to improved classification accuracy. This was achieved by replicating each sample in the original training set, increasing the number of training samples from N to $2N$. The augmented dataset was subsequently employed in the k -fold partitioning utilized within the bagging framework. As a result, individual samples could appear twice within the same fold. The duplication introduced additional variability among the training subsets used for model generation, not only due to the inherent sample differences across folds but also due to differences in sample distribution. This increased diversity among the base models contributed to enhanced ensemble performance.

2.7 Two-Phase Cosine Annealing

We introduce an enhanced schedule based on cosine annealing [35]. In this approach, The initial learning rate is set to 0.001. As training progresses, the learning rate undergoes a gradual decay during early epochs, accelerates its descent in the mid-phase, and eventually stabilizes at a lower value. This dynamic adjustment aligns with the varying requirements of different optimization phases: large learning rates promote global exploration in early stages, while smaller rates facilitate fine-tuning in later epochs. Under this setup, the use of a separate validation set is no longer required. Instead, the model parameters from the final training epoch are adopted as the final result. After reaching the minimum learning rate, the schedule resets it back to the initial value (0.001), initiating a new cycle of cosine annealing. This restart mechanism enables the optimizer to potentially escape local minima and continue effective exploration. The complete learning rate schedule is detailed in Algorithm 2.

Algorithm 2 Learning rate schedule of ATCNet-WS

Input: initial model M
Output: trained model M

```

1: learning rate  $lr \leftarrow 0.001$ 
2:  $lr\_min1 \leftarrow 0.0001$ 
3:  $lr\_max1 \leftarrow 0.001$ 
4:  $cycle1 \leftarrow 500$ 
5: for  $epoch = 1$  to  $cycle1$  do
6:   train  $M$  for one epoch with learning rate  $lr$ 
7:    $lr \leftarrow lr\_min1 + \frac{lr\_max1 - lr\_min1}{2} (1 + \cos \frac{epoch \cdot \pi}{cycle1})$ 
8: end for
9:
10: learning rate  $lr \leftarrow 0.001$ 
11:  $lr\_min2 \leftarrow 0.00002$ 
12:  $lr\_max2 \leftarrow 0.001$ 
13:  $cycle2 \leftarrow 2 * cycle1$ 
14: for  $epoch = 1$  to  $cycle2$  do
15:   train  $M$  for one epoch with learning rate  $lr$ 
16:    $lr \leftarrow lr\_min2 + \frac{lr\_max2 - lr\_min2}{2} (1 + \cos \frac{epoch \cdot \pi}{cycle2})$ 
17: end for
18:
19: return  $M$ 

```

3 Experiments

3.1 Evaluation Setup

We adopt a subject-dependent evaluation approach, wherein a separate model is trained individually for each subject. The training and testing set configurations for the BCI Competition IV 2a and 2b datasets follow the settings described in Section II-A. All model hyperparameters remain consistent across all subjects in both datasets.

Due to the stochasticity in neural network training, we report the final classification accuracy for each subject as the average of three independent training runs. This procedure is applied uniformly to ATCNet-WS as well as all models from the literature. All training is performed on a NVIDIA RTX 4090 GPU.

In the evaluation procedures of prior works [20], [21], [27], based on the authors' released source codes, we found that the reported final accuracies in those studies were obtained by using the test set as a validation set—selecting the model corresponding to the epoch with the highest test accuracy as the final result. This is not a valid evaluation approach, as it compromises the integrity of the evaluation by allowing test data to influence model training.

To ensure a fair and unbiased comparison, we strictly exclude test data from the training process in all our experiments. For models lacking a predefined strategy for selecting the final model—unless the original source code explicitly avoids using the test set during training—we adopt a validation-based approach: the original training set is split into training and validation sets in a 4:1 ratio. The model is trained on the training subset, and the epoch yielding the lowest validation loss is used as the final result.

3.2 Comparison to Literature Models

The literature models included in our evaluation for comparison are configured as follows:

1. ATCNet[27], TCNet[20], TCNet-Fusion[21], MBEEG-SENet[22]: The implementation of the four models was based on the publicly available code at <https://github.com/Altaheri/EEG-ATCNet>. As the original implementation utilized the test set as a validation set, we modified the code to adopt the validation-based approach (detailed in Section III-A). The initial learning rate was set to 0.001 and was reduced by a factor of 0.1 if the validation loss did not improve over 20 consecutive epochs.
2. M-ShallowConvNet[23]: Source code is available at <https://github.com/comojin1994/m-shallowconvnet>. As the original implementation does not utilize test set data during training, we retain all default settings without modification.

3. DB-ATCNet[28]: Source code is available at <https://github.com/zk-xju/DB-ATCNet>. As the original implementation utilized the test set as a validation set, we modified the training process to employ the same validation-based approach and learning rate schedule used in ATCNet. All other components of the original code remain unchanged.
4. MSVTNet[26]: Source code is available at <https://github.com/SheepTAO/MSVTNet>. Since the training process does not involve test data, the original configuration is maintained.

The classification accuracies of all models on datasets 2a and 2b are reported in Table 1 and Table 2, respectively. It can be observed that, with the exception of M-ShallowConvNet and MSVTNet, most literature models experience a noticeable performance drop compared to the results reported in their original publications. On dataset 2a, ATCNet-WS achieves the highest accuracy across nearly all subjects, with the only exception being Subject 9, where M-ShallowConvNet performs slightly better. On average across all subjects, ATCNet-WS significantly outperforms the other models. Similarly, on Dataset 2b, ATCNet-WS attains the highest average accuracy; however, the performance gap is narrower. This is due to that 2b is a two-class dataset, where the lowest classification accuracy is already at 50%, resulting in reduced variability among different methods.

Table 1: Accuracy(%) comparison to literature models on BCI Competition IV dataset 2a

models	Sub1	Sub2	Sub3	Sub4	Sub5	Sub6	Sub7	Sub8	Sub9	Average
ATCNet[27]	84.12	68.84	92.52	82.36	76.02	71.32	90.07	84.10	85.83	81.69
TCNet[20]	67.56	46.18	79.07	52.03	67.40	52.36	76.25	72.98	70.55	64.93
TCNet-Fusion[21]	73.68	45.92	85.06	58.65	66.11	55.85	82.91	78.65	75.25	69.12
M-ShallowConvNet[23]	90.05	57.29	94.68	80.79	79.28	67.94	92.48	83.56	87.96	81.56
MBEEG-SENet[22]	77.94	57.18	90.63	62.71	53.43	56.64	84.54	81.76	77.15	71.33
DB-ATCNet[28]	82.41	59.21	90.44	75.21	69.72	66.81	87.78	84.81	83.98	77.82
MSVTNet[26]	83.10	55.02	91.44	74.32	72.43	68.12	87.52	83.10	83.82	77.65
ATCNet-WS	91.55	73.03	95.72	90.97	82.29	79.86	94.10	88.19	87.85	87.06

"Subx" refers to the x-th subject in the BCI Competition IV-2a dataset. "Average" represents the mean accuracy across 9 subjects. The highest accuracy for each subject and the average accuracy are highlighted in bold.

Table 2: Accuracy(%) comparison to literature models on BCI Competition IV dataset 2b

models	Sub1	Sub2	Sub3	Sub4	Sub5	Sub6	Sub7	Sub8	Sub9	Average
ATCNet[27]	77.04	72.21	84.29	96.58	90.33	85.54	91.31	92.15	87.06	86.28
TCNet[20]	73.60	67.79	75.67	97.13	92.38	76.38	89.69	92.06	84.94	83.29
TCNet-Fusion[21]	73.52	68.67	81.63	96.40	93.38	84.94	90.88	91.77	85.35	85.17
M-ShallowConvNet[23]	78.96	68.68	81.04	94.37	98.86	85.62	91.98	91.67	83.33	86.06
MBEEG-SENet[22]	76.77	67.64	72.23	96.44	91.63	85.83	90.77	92.83	87.27	84.60
DB-ATCNet[28]	73.17	69.83	81.65	97.40	92.60	83.48	90.96	86.65	84.67	84.49
MSVTNet[26]	77.81	70.10	84.58	96.23	94.92	85.52	91.79	93.31	87.69	86.88
ATCNet-WS	78.44	72.86	85.00	97.50	96.56	87.81	92.19	93.12	87.19	87.85

"Subx" refers to the x-th subject in the BCI Competition IV-2b dataset. "Average" represents the mean accuracy across 9 subjects. The highest accuracy for each subject and the average accuracy are highlighted in bold.

3.3 Ablation Analysis on Weight Sharing and Bagging

Table 3 presents the classification accuracies of ATCNet-WS after independently removing the weight sharing and bagging components on the BCI Competition IV 2a and 2b datasets. The ablation of weight sharing was implemented by replacing the temporal module in ATCNet-WS with the version described in Algorithm 3, while keeping all other components unchanged. In this version, 4 dilated temporal convolution layers (conv1, conv2, conv3, conv4) are defined and each used once in the blocks. To ablate the bagging strategy, a single model was trained per subject using the entire original training set, without performing the 5-fold partitioning and ensemble employed in the original bagging approach.

As shown in Table 3, the removal of either weight sharing or bagging leads to a significant reduction in classification accuracy across both datasets. These results highlight the critical contributions of both strategies to the overall performance of the ATCNet-WS framework.

Algorithm 3 Temporal module without weight sharing

Input: tensor x
Output: tensor y

```

1:  $conv1 \leftarrow \text{conv1D\_layer}(\text{dilation\_rate}=1)$ 
2:  $conv2 \leftarrow \text{conv1D\_layer}(\text{dilation\_rate}=1)$ 
3:  $z \leftarrow x$ 
4:  $x \leftarrow conv1(x)$ 
5:  $x \leftarrow \text{BatchNorm}(x)$ 
6:  $x \leftarrow \text{ELU}(x)$ 
7:  $x \leftarrow \text{Dropout}(x)$ 
8:  $x \leftarrow conv2(x)$ 
9:  $x \leftarrow \text{BatchNorm}(x)$ 
10:  $x \leftarrow \text{ELU}(x)$ 
11:  $x \leftarrow \text{Dropout}(x)$ 
12:  $x \leftarrow x + z$ 
13:  $x \leftarrow \text{ELU}(x)$ 
14:
15:  $conv3 \leftarrow \text{conv1D\_layer}(\text{dilation\_rate}=2)$ 
16:  $conv4 \leftarrow \text{conv1D\_layer}(\text{dilation\_rate}=2)$ 
17:  $z \leftarrow x$ 
18:  $x \leftarrow conv3(x)$ 
19:  $x \leftarrow \text{BatchNorm}(x)$ 
20:  $x \leftarrow \text{RELU}(x)$ 
21:  $x \leftarrow \text{Dropout}(x)$ 
22:  $x \leftarrow conv4(x)$ 
23:  $x \leftarrow \text{BatchNorm}(x)$ 
24:  $x \leftarrow \text{RELU}(x)$ 
25:  $x \leftarrow \text{Dropout}(x)$ 
26:  $y \leftarrow x + z$ 
27:  $y \leftarrow \text{ELU}(x)$ 
28:
29: return  $y$ 

```

Table 3: Accuracy(%) after removing weight sharing and bagging

methods	2a	2b
Original ATCNet-WS	87.06	87.85
Weight sharing removed	85.68	87.02
Bagging removed	85.48	86.72
"2a" and "2b" represent the mean accuracy across 9 subjects for datasets 2a and 2b, respectively.		

3.4 Ablation Analysis on Sample Replication

We investigated the effect of sample replication through two ablation settings. In the first setting, training was performed directly on the original training set without sample replication. In the second setting, sample replication was removed and the total number of training epochs was doubled. Specifically, this was implemented by modifying cycle1 in Algorithm 2 to 1000.

The classification accuracies obtained under these two configurations on the 2a and 2b datasets are summarized in Table 4. The original ATCNet-WS model, which includes sample replication, achieved the highest accuracy. The configuration with doubled training epochs outperformed the setting using the original dataset with the original number of epochs but underperformed compared to the original ATCNet-WS.

These results suggest that the benefits of sample replication extend beyond merely increasing the total number of training epochs. Sample replication introduces variability not only in the composition of training samples across folds but also in the sample distribution within each fold. This enhances the diversity among individual models in the ensemble, thereby improving the overall performance of the model after bagging.

Table 4: Accuracy(%) after removing sample replication

methods	2a	2b
Original ATCNet-WS	87.06	87.85
Sample replication removed	86.27	86.75
Sample replication removed and double epoch	85.69	87.13

"2a" and "2b" represent the mean accuracy across 9 subjects for datasets 2a and 2b, respectively.

3.5 Ablation Analysis on Signal Extention

We provide three configurations for signal extension based on the number of windows employed in the sliding window block: 5, 6, and 7. The configuration with 6 windows corresponds to the current ATCNet-WS approach. In the 7-window setting, the input signal length remains 1250, the same as in the 6-window case; however, all seven sub-matrices are utilized for generating the branches, in contrast to the 6-window approach which discards the first sub-matrix.

In the 5-window setting, the input signal has a length of 1125 time points, which corresponds to a 4-second segment after the cue, combined with a 0.5-second pre-cue segment. Within the convolutional block, two average pooling operations with pool sizes of 8 and 7 are applied, respectively, reducing the length of the signal by a factor of approximately 56. This results in a temporal length of 20 time points as the input to the sliding window block. Applying a sliding window of length 16 under this configuration yields five sub-matrices, and thus, five corresponding branches.

Table 5 summarizes the classification accuracy and training time for each configuration on the BCI Competition IV 2a and 2b datasets. Due to the extensive computational cost of training ensembles with bagging, we report single-model performance without bagging. Results show that increasing the number of windows from 5 to 6 leads to improved accuracy, indicating the benefit of the extended temporal context. However, further increasing the window count to 7 yields negligible accuracy improvement while significantly increasing training time due to the addition of an extra network branch.

Based on this trade-off between accuracy and computational cost, we adopt the 6-window configuration in our final model.

Table 5: Accuracy(%) and training time(seconds) of different window numbers

methods	2a accuracy	2b accuracy	2a time	2b time
5 windows	85.18	86.54	546	449
6 windows	85.48	86.72	605	495
7 windows	85.50	86.52	665	541

"2a accuracy" and "2b accuracy" represent the mean accuracy across 9 subjects for datasets 2a and 2b, respectively. "2a time" and "2b time" represent the mean training time across 9 subjects for datasets 2a and 2b, respectively. Due to the extensive computational cost of training ensembles with bagging, we report single-model performance without bagging.

3.6 Ablation Analysis on Two-Phase Cosine Annealing

We provide four configurations for the ablation analysis on two-phase cosine annealing:

1. Two-phase cosine annealing: This is the current schedule used in ATCNet-WS, as detailed in Algorithm 2.
2. Reduce learning rate on plateau: In this setting, the initial learning rate is set to 0.001. The learning rate is reduced by a factor of 0.1 whenever the validation loss failed to improve for more than 20 consecutive epochs.

3. Fixed learning rate: In this setting, the model is trained with a constant learning rate of 0.001 for 1500 epochs. The model after the last epoch is used as the final result. Unlike the validation-based schedule, this configuration allows the model to utilize the entire training dataset, thereby potentially benefiting from increased sample number.
4. One-phase cosine annealing: This schedule applies cosine annealing over 1500 epochs, starting from a learning rate of 0.001 and decaying to 0.00002, without learning rate reset.

The classification accuracies achieved by these four configurations on datasets 2a and 2b are summarized in Table 6. Due to the extensive computational cost of training ensembles with bagging, we report single-model performance without bagging. Among all configurations, the two-phase cosine annealing yielded the highest accuracy. This was followed by the one-phase cosine annealing, then the reduce learning rate on plateau, and finally the fixed learning rate approach.

These results highlight the importance of both cosine annealing and learning rate reset in achieving optimal performance. A gradually decaying learning rate appears crucial. Moreover, scheduling strategies that do not rely on a validation set enable full utilization of the training data, potentially offering a performance advantage.

Table 6: Accuracy(%) of different learning rate schedules

methods	2a	2b
Two-phase cosine annealing	85.48	86.72
One-phase cosine annealing	85.00	86.10
Reduce learning rate on plateau	82.38	85.72
Fixed learning rate	76.27	85.70

"2a" and "2b" represent the mean accuracy across 9 subjects for datasets 2a and 2b, respectively. Due to the extensive computational cost of training ensembles with bagging, we report single-model performance without bagging.

3.7 Impact of Epoch Number

This section investigates the influence of the total number of training epochs on classification performance. We evaluate five different total epoch number settings: 300, 400, 800, 1500, and 2100. These correspond to setting *cycle1* in Algorithm 3 to 100, 200, 400, 500, and 700, respectively. The resulting classification accuracies are illustrated in Fig. 4, where the red and blue lines represent results on dataset 2a and 2b, respectively. Due to the extensive computational cost of training ensembles with bagging, we report single-model performance without bagging.

As shown, the number of epochs has a more pronounced effect on the accuracy for dataset 2a, with the highest performance achieved at 1500 epochs. In contrast, dataset 2b exhibits relatively minor variations in accuracy across the different epoch settings. Considering the performance trends observed across both datasets, we selected 1500 as the optimal total number of training epochs.

4 CONCLUSION

In this study, we proposed ATCNet-WS, a novel architecture for EEG-based MI classification. The model comprises four blocks. First, the Convolutional Block extracts rich spatiotemporal features directly from raw EEG signals. Second, the Sliding Window Block partitions these features into multiple sub-matrices, each forming an independent branch, thereby enhancing the network’s width and representational capacity. Third, the ATC Block integrates information across the full temporal sequence through a combination of multi-head self-attention (MSA) and multi-layer dilated temporal convolutions. Importantly, weight sharing across dilated convolutional layers reduces overfitting and improves generalization. Fourth, the Decision Block aggregates outputs from all branches to produce the final classification. Furthermore, we employed a combination of bagging and sample replication, which significantly improved both accuracy and robustness. The adoption of a two-phase cosine annealing learning rate schedule outperformed conventional validation-based schedule.

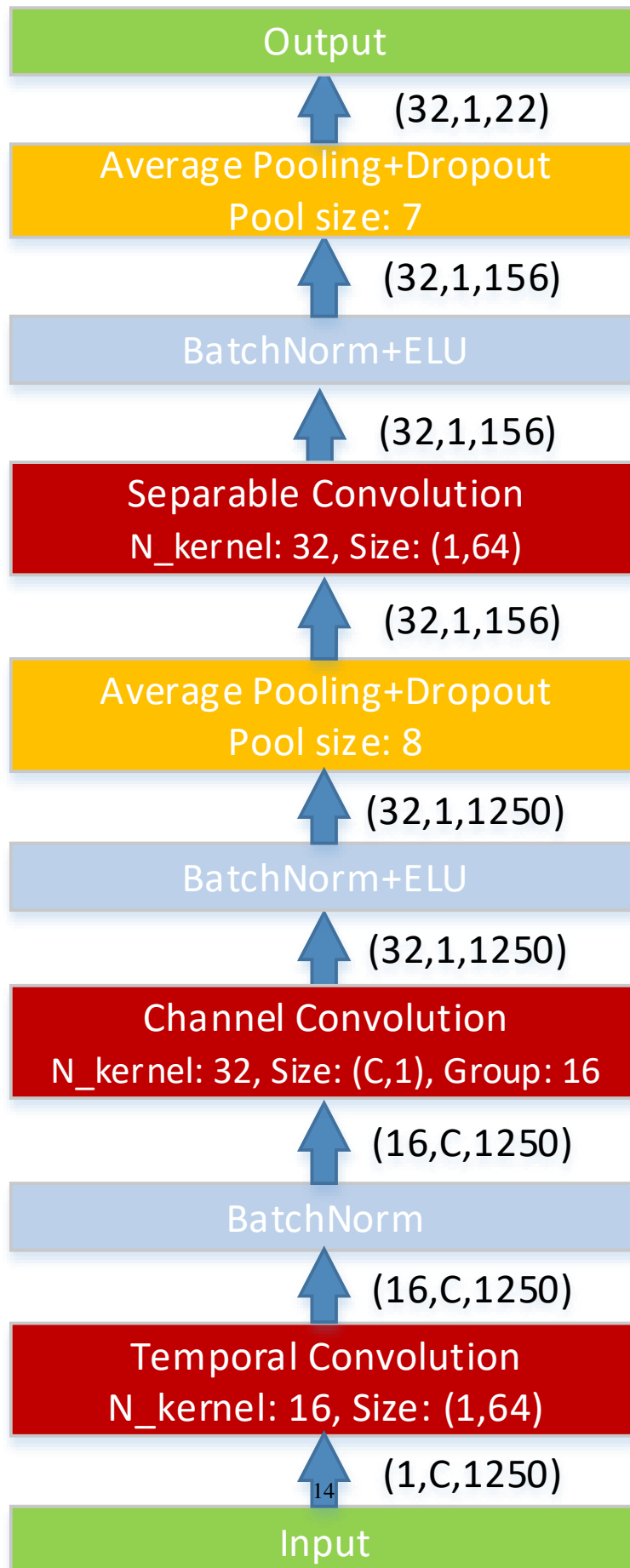
Extensive experiments on the BCI Competition IV 2a and 2b datasets demonstrate the effectiveness of ATCNet-WS, achieving classification accuracies of 87.06% and 87.85%, respectively—substantially outperforming existing methods reported in the literature. Comprehensive ablation studies confirm the individual contributions of each strategy.

Future work will explore novel attention mechanisms and loss function designs, which we believe hold substantial promise for further improving MI classification.

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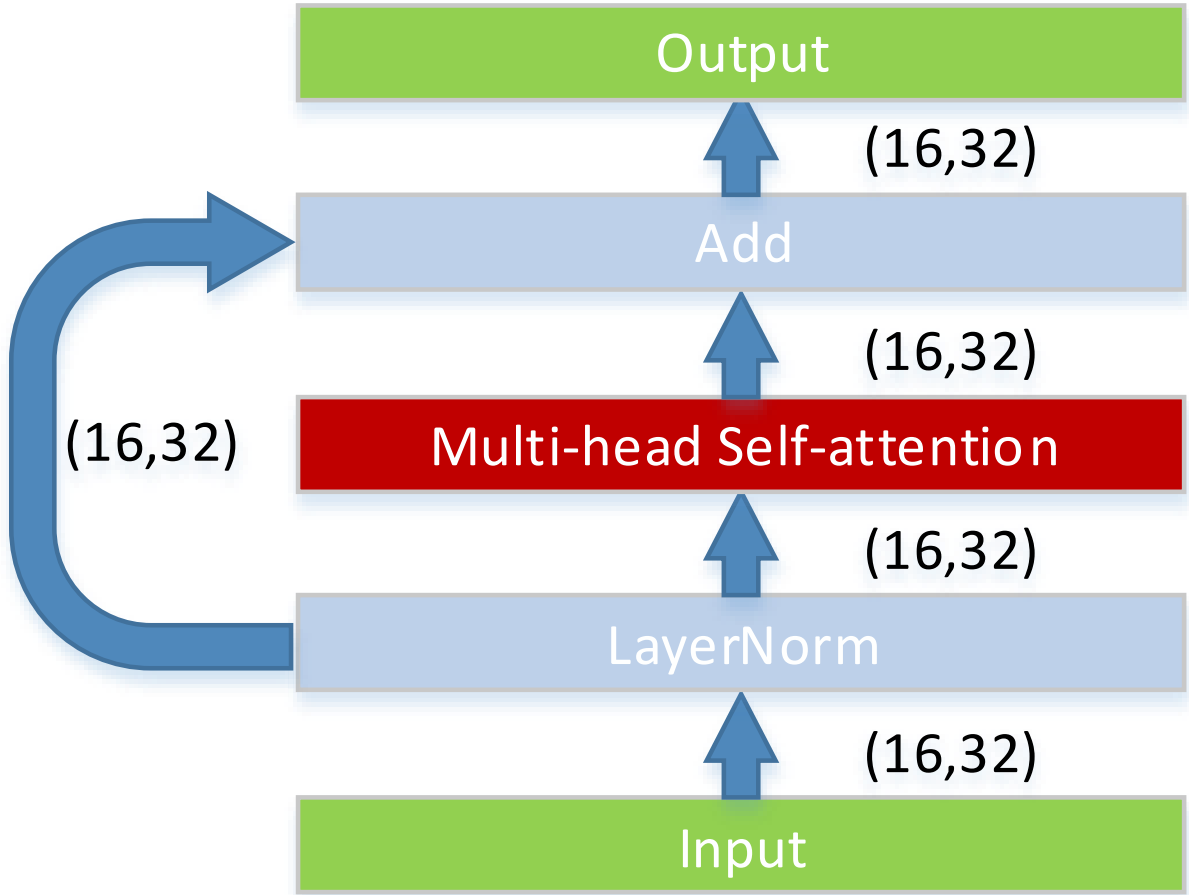


Figure 3: Structure of the attention module. The numbers on the right side of the arrows indicate the size of the tensor input to the next layer. The MSA layer is implemented using TensorFlow’s MultiHeadAttention, with 2 heads and a key dimension of 8.

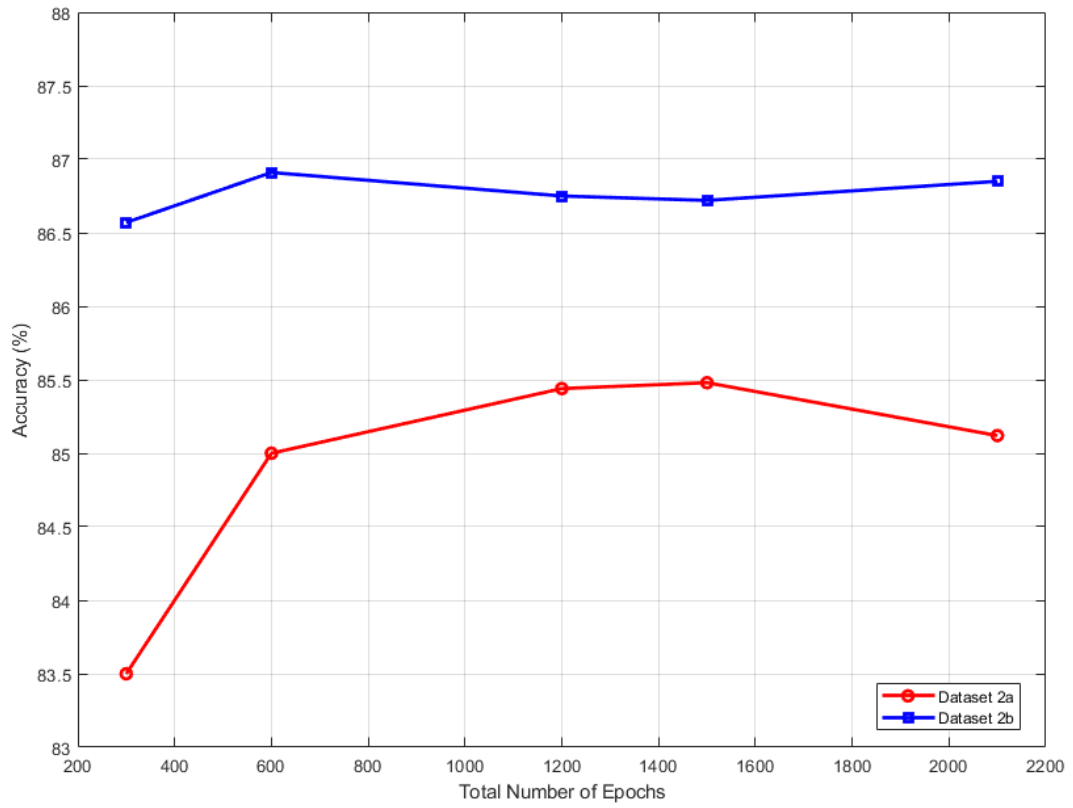


Figure 4: Impact of total epoch number on accuracy. "Accuracy" represent the mean accuracy across 9 subjects for dataset 2a or 2b.